SignSernade: Your Voice in Signs

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Abstract—"SignSerenade: Your Voice in Signs" is a platform that bridges communication barriers for Deaf and hard-of-hearing communities. It provides real-time American Sign Language (ASL) translation using advanced deep learning models like YOLOv11 and MediaPipe, integrated with the Word Level ASL (WLASL) dataset. Key features include bidirectional translation of ASL signs and an interactive learning module with tutorials, quizzes, and real-time gesture feedback. The system finds applications in education, healthcare, and customer service, promoting inclusivity. Future developments aim to improve recognition accuracy, include facial expression analysis, and enable offline functionality, fostering a more inclusive society.

Keywords—ASL, deep learning, real-time translation, inclusivity, gesture recognition.

I. INTRODUCTION

Millions of individuals in the Deaf and hard-of-hearing community face significant communication challenges due to the lack of effective and accessible tools for American Sign Language (ASL) translation and learning. Traditional ASL learning systems often lack real-time processing capabilities and fail to provide meaningful feedback, which limits the engagement and effectiveness of these platforms.

"SignSerenade" addresses these issues by utilizing the WLASL dataset to enable real-time ASL-to-text and text-to-ASL translation. The system offers an interactive learning experience, including video tutorials, quizzes, and instant feedback, making ASL education both accessible and engaging for users of all ages and backgrounds.

II. DATASET

The success of machine learning models for sign language recognition largely depends on the quality and diversity of the datasets. This project utilizes two key datasets for gesture detection and text-to-sign translation.

A. Custom ASL Gesture Dataset

The **Custom ASL Gesture Dataset** was designed to train the YOLOv11 model for ASL gesture recognition. It contains images and videos featuring hand and face landmarks that are crucial for accurate gesture classification. To minimize environmental interference, a plain black background is used in all images. Each image corresponds to a specific ASL word or letter, ensuring a clear mapping between the gesture and its meaning. The dataset is split into training, validation, and test sets, and various data augmentation techniques, such as rotation, zooming, and flipping, are applied to enhance model robustness by simulating a variety of real-world conditions.

B. WLASL Dataset

The **WLASL (Word-Level American Sign Language) Dataset** is employed for text-to-sign translation, containing high-quality videos of individuals performing ASL signs for specific words. Each video represents an individual ASL word, providing valuable data for mapping text input to corresponding sign language gestures. The dataset includes a standardized vocabulary of ASL words, ensuring consistency and clarity in translation. The videos feature diverse signers which promote generalization and improve the model's ability to recognize signs from different users and settings.

C. Challenges with the Dataset

During the creation and preprocessing of the datasets, several challenges arose. The **Custom ASL Gesture Dataset** faced limitations in diversity, particularly in hand shapes and lighting conditions. To address these limitations, data augmentation techniques were applied to simulate various hand shapes and lighting conditions. The **WLASL Dataset** contained some videos with low resolution, which was addressed by using frame interpolation to improve video quality. Additionally, ASL signs can vary regionally and individually, which poses a challenge for model generalization. To tackle this, the model was trained on a

wide range of signers to ensure that it could recognize variations in sign language across different individuals.

D. Dataset Summary

The **Custom ASL Gesture Dataset** is used primarily for hand gesture detection and landmark classification, with augmented images to improve the model's training process. On the other hand, the **WLASL Dataset** provides high-quality, video-based data for text-to-sign translation, with a standardized ASL vocabulary to ensure consistency and reliability in gesture recognition. These datasets serve as foundational resources for developing an accurate and robust system for sign language recognition and translation.

III. PREPROCESSING

Normalization, quality enhancement, and labeling are crucial preprocessing steps in preparing the dataset for YOLOv11 model training.

A. Normalization:

Normalization involves standardizing the image dimensions by resizing all images to a consistent size, such as 416x416 pixels. This ensures uniformity across the dataset, allowing the YOLOv11 model to process inputs effectively without variation in dimensions. Additionally, the coordinates of facial and hand landmarks are normalized relative to the image dimensions, ensuring compatibility across varying resolutions and maintaining consistency in landmark representation.

B. Quality Enhancement

Quality enhancement improves the dataset's visual clarity by applying techniques like contrast adjustment to emphasize landmarks, making them more distinguishable for detection. Denoising filters are used to remove any unnecessary artifacts or visual noise, ensuring that the landmarks remain the primary focus of the image.

C. Labeling

Labeling assigns specific identifiers to each image to map it to the corresponding word or letter. If landmarks are overlaid on images, bounding boxes are created around the regions of interest, such as the hands and face, to define the detection zones. Additionally, textual labels such as "HELLO" or individual letters like "A" are attached to the images to represent the corresponding word or letter. These preprocessing steps collectively ensure the dataset is optimized for training, enhancing model performance and accuracy.

IV. ABSTRACT DESIGN

The abstract design provides a high-level view of the "SignSerenade" system, outlining the core components and their interactions within the system's architecture. This design is essential for visualizing how the system functions as a whole, showcasing the various modules and their respective roles in delivering the desired outcomes.

The architecture is structured around two primary components: **gesture recognition** and **text-to-sign translation**. The **gesture recognition** module leverages the YOLOv11 model, which is highly efficient for detecting and classifying American Sign Language (ASL) gestures in real time. The model processes images or video frames to identify key hand and face landmarks, allowing the system to translate these gestures immediately into their corresponding text form. This functionality makes the system capable of performing real-time ASL-to-text translation, which is beneficial for users who wish to communicate with text output based on their sign language input.

On the other hand, the **text-to-sign translation** component of the system is driven by the WLASL dataset, which includes video clips of various ASL signs. When a user enters text, the system retrieves the corresponding ASL video from the WLASL dataset, displaying the video to visually represent the translated sign. This allows users to learn the correct sign language gestures associated with the text, thus facilitating text-to-sign translation. Together, these components work in tandem to offer users an interactive and real-time ASL learning experience, promoting both sign language recognition and understanding.

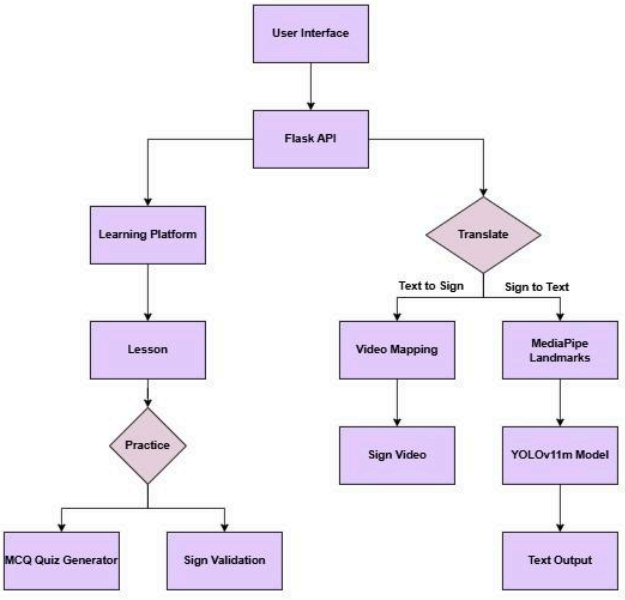


Fig 1. Architectural Design

VI. FUNCTIONAL DESIGN

The functional design of the "SignSerenade" system dives deeper into the operational workflows, specifying the processes involved when handling user input and generating output. This design is critical in defining the way data flows through the system, ensuring that each task is processed efficiently and effectively.

The system operates based on two types of input: **ASL gestures** and **text**. When the system receives an **ASL gesture** as input, it uses the YOLOv11 model to detect the gesture by identifying hand and face landmarks. These landmarks are then classified and mapped to the corresponding text, allowing the system to translate the gesture into written language. The output is displayed immediately, providing real-time translation for users.

For **text input**, the system performs a reverse operation. It uses the WLASL dataset to find the appropriate ASL sign video that corresponds to the entered text. This video is then presented to the user, visually demonstrating the ASL gesture for the word or phrase in question. Additionally, the system includes an **interactive learning module** that enhances the learning experience. This module offers step-by-step tutorials that guide users in performing ASL gestures and quizzes that assess their learning progress. The system evaluates users' performance and provides instant feedback, allowing them to adjust their gestures and improve over time.

Thus, the functional design enables seamless interaction between input, processing, and output, while also incorporating features that foster learning, such as tutorials, quizzes, and real-time feedback.

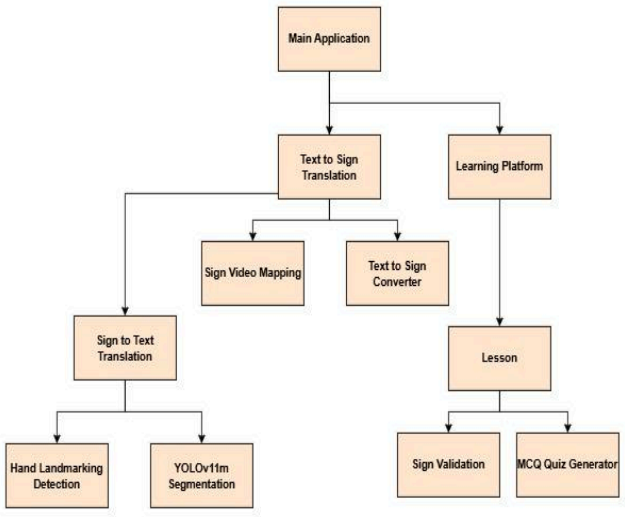


Fig 2. Modular Design diagram

VII. SEQUENCE DESIGN

The sequence design defines the step-by-step process the system follows to handle user inputs and generate corresponding outputs, ensuring that each operation occurs in a logical and efficient order to enhance the user experience.

The process begins when the system receives input from the user. The system first determines the type of input: whether it is an ASL gesture or text. If the input is an ASL gesture, the system captures the video frames and passes them through the YOLOv11 model. The model processes the frames to identify key landmarks on the hand and face and classifies the gesture accordingly. Once the gesture is classified, the system generates the corresponding text representation and displays it to the user as the output.

In the case of text input, the system performs a search through the WLASL dataset to find the corresponding ASL sign video. Upon locating the relevant video, the system displays it to the user, showcasing the correct ASL gesture for the given word or phrase. This real-time translation of text into ASL ensures that users can visualize the corresponding gesture instantly.

After presenting the output, the system provides real-time feedback based on the user’s performance. For gesture input, the system checks whether the recognized gesture matches the expected text and provides visual feedback to indicate its accuracy. Correct gestures are highlighted in green, while incorrect ones are marked in red, signaling the need for adjustment. For learning activities, such as tutorials and quizzes, the system provides corrective feedback, encouraging the user to adjust their gestures based on their performance. This immediate feedback loop helps users refine their skills and improve gesture recognition.

The system operates in a continuous learning loop, constantly monitoring the user’s progress. As users practice, they are guided toward more accurate gesture recognition through ongoing feedback. This adaptive learning process ensures that users remain engaged and are consistently improving their ASL proficiency. For exercises like tutorials and quizzes, the system encourages repetition, allowing users to practice gestures until they master them.

Overall, the sequence design ensures that the system processes inputs and outputs in a structured manner, providing users with immediate feedback, supporting continuous learning, and enhancing their ASL recognition skills. The design ensures that the system remains responsive

and adaptive, facilitating an effective and enjoyable learning experience.

VIII. PRESENTATION DESIGN

The presentation design focuses on the visual aspects and user interface (UI) of the "SignSerenade" system, emphasizing clarity, usability, and accessibility. The goal is to create an interface that is intuitive and engaging for users, making the learning process enjoyable and effective.

The user interface is designed with simplicity in mind, avoiding clutter and unnecessary distractions. Key components, such as the **gesture recognition screen**, **text-to-sign translation interface**, and **interactive learning modules**, are prominently displayed for easy navigation. The design uses a clean layout with large, easy-to-read text and icons that guide the user through the different features of the system.

When translating **text into ASL**, the system shows high-quality video clips sourced from the WLASL dataset. These videos are carefully chosen to ensure clarity, with close-up views of the singer's hand movements and facial expressions. Annotations are included within the video to highlight critical movements, helping users understand the correct execution of each ASL gesture. Moreover, the feedback system is visual, providing immediate color-coded indicators that show whether a gesture was recognized accurately. Green indicates correct recognition, while red suggests a need for adjustment, prompting the user to refine their gestures.

This visual feedback mechanism is integral to the learning process, as it helps users track their progress and make improvements based on real-time assessments. Overall, the presentation design ensures that the system is not only functional but also engaging, making the ASL learning experience more interactive and accessible

IX. TRAINING

In this section, we discuss the process of training the models used for gesture recognition and sign language translation. The training phase focuses on optimizing the performance of the **YOLOv11** model for accurate gesture detection and the utilization of the **WLASL dataset** for translating text to ASL gestures. Both models are fine-tuned to handle various challenges such as dataset diversity, hand pose variation, and environmental factors.

A. Model Architecture and Selection

For **Text-to-Sign translation**, the **WLASL dataset** is utilized in combination with a video retrieval system. When a user inputs text, the system maps the text to the corresponding ASL video from the WLASL dataset, providing real-time sign language demonstrations. This method ensures accurate representation of ASL gestures through pre-recorded videos associated with specific text words.

For **gesture detection**, the **YOLOv11** model was selected due to its efficiency and performance in real-time object detection tasks. YOLOv11 is designed to detect and classify objects, making it ideal for recognizing ASL signs based on the landmarks extracted from input images or video frames. The model processes key features such as hand and face landmarks using a custom dataset, enabling it to differentiate between various ASL signs accurately.

B. Training Process

The **YOLOv11 model** is trained using a **supervised learning approach**. The labeled dataset, consisting of images of ASL gestures, is divided into **training**, **validation**, and **test** sets. Initially, the model is initialized with **pre-trained weights**, which are then fine-tuned on the ASL dataset to optimize performance. During training, the model learns to detect hand and face landmarks and classify ASL gestures. Hyperparameters such as learning rate, batch size, and the number of epochs are carefully tuned to maximize model accuracy. Additionally, **data augmentation techniques** (e.g., flipping, rotation, and zooming) are employed to improve the model's robustness and generalization to various input scenarios.

For the **Text-to-Sign** translation, the **WLASL dataset** does not require a traditional training process. Instead, the dataset is used to **map text input** to the corresponding ASL videos. These videos represent the sign language gestures for specific words, providing an accurate translation from text to ASL gestures.

C. Model Evaluation and Fine-Tuning

Following the training of the **YOLOv11 model**, its performance is evaluated on a separate test set, where key metrics such as **accuracy**, **precision**, **recall**, and **F1-score** are calculated. These metrics provide a clear picture of the model's ability to classify ASL signs correctly. Based on these evaluation results, the model is fine-tuned to improve its performance, particularly in challenging environments with varying lighting conditions, hand poses, or background interference.

Additionally, **sign-to-text translation** accuracy is assessed through **real-world testing**, where the model processes user gestures and compares the output with expected results. The feedback obtained from this testing is used to further refine both gesture detection and sign recognition, ensuring that the system performs effectively in diverse scenarios.

D. Challenges and Solutions

During the training phase, several challenges were encountered, such as **limited dataset diversity**, variations in **hand shapes**, and **environmental factors** like poor lighting and background noise. To mitigate these issues, various **data augmentation techniques** were applied, and adjustments were made to the model architecture to handle a broader range of inputs. The system was tested in various environments to ensure robustness, and fine-tuning was carried out based on user feedback to improve the model's performance in real-world settings. These steps helped enhance the model's generalization capability and ensured reliable ASL recognition across different conditions.

The image presents four plots summarizing the training and validation metrics over 15 epochs. The **Train Loss** (top left) plot shows a steady decrease, indicating effective learning, with a smoothed trend confirming this. The **Validation Loss** (top right) also declines, reflecting the model's ability to generalize well, with both "results" and "smooth" lines following a similar downward trajectory. The **Accuracy Top-1** (bottom left) plot shows steady improvement, reaching nearly 100% by the 15th epoch, while the **Accuracy Top-5** (bottom right) quickly nears 100% in the early epochs, suggesting the model captures key patterns early in training. These metrics highlight the model's strong performance in learning and generalizing to new data.

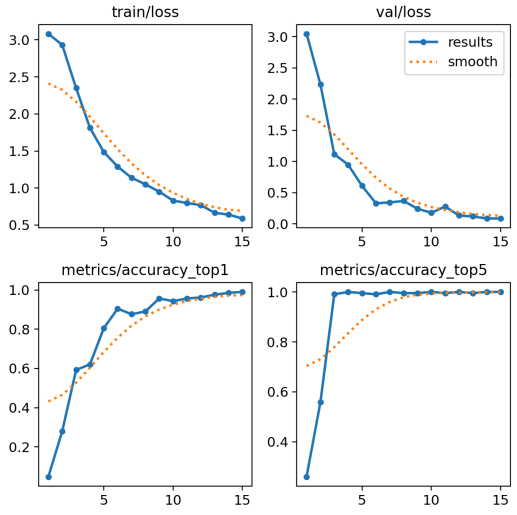


Fig 3. Model training and validation metrics

X. TESTING

The testing process for the SignSerenade platform was designed to evaluate its performance, accuracy, and robustness in different scenarios. Multiple test cases were created for each module, focusing on functionality, edge cases, and integration. The results demonstrated both successes and areas for improvement.

A. Sign Language Detection Module

Objective: Evaluate the ability of the system to detect and translate ASL gestures into text under various conditions.

Table 1. Test Scenarios for the Sign Language Detection Module

Test Case	Input	Expected Output	Result
Single sign detection	The user performs a "hello" gesture	Model outputs "hello"	Pass
Variations in gesture	Slight hand position/angle change	The model detects the correct word	Fail
Invalid Input	Random hand gesture	Model outputs "No match"	Fail
Background interference	Gesture in a noisy background	Accurate gesture detection	Pass
Real-time detection	Continuous video input	Timely word detection	Pass

B. Text-to-Sign Mapping Module

Objective: Assess how accurately text inputs are converted into corresponding ASL video outputs.

Table 2. Test Scenarios forText-to-Sign Mapping with WLASL Dataset

Test Case	Input	Expected Output	Result
Valid word mapping	"Hello"	Displays WLASL video for 'hello.mp4'	Pass
Invalid word input	"XYZ"	Outputs "Word not found" error	Pass
Case sensitivity	"Hello"	Correct mapping despite the case	Pass
Large input batch	Multiple words	Accurate mapping for each word	Pass

API response time	"Thank you"	Video retrieved within 2-3 seconds	Pass

C. Interactive Learning Platform

The learning platform, which includes a quiz module, provided an engaging and effective educational experience. The system displayed WLASL videos correctly and confirmed accurate answers during quizzes. Incorrect answers were promptly flagged, ensuring that users received meaningful feedback. Randomization of quiz options and videos added variety, enhancing the user experience. Furthermore, the system handled edge cases like missing video references gracefully by displaying appropriate error messages. Gesture validation within the learning platform was generally accurate for common signs like "thank you." However, some challenges arose with incorrect or untrained gestures, where the system occasionally failed to recognize invalid inputs.

D. Integration Testing

End-to-end testing confirmed seamless functionality across all system components. The platform successfully detected signs and mapped text inputs to corresponding ASL videos, ensuring a smooth user experience. Quiz interaction flows, including validation feedback, performed consistently, reinforcing the educational aspect of the platform. Real-time performance in sign-to-text and text-to-sign conversions demonstrated the system's robustness and usability for practical applications.

XI. CONCLUSION

The "SignSerenade: Your Voice in Signs" project provides a groundbreaking solution to address communication challenges for the Deaf and hard-of-hearing communities. Utilizing the WLASL dataset and advanced deep learning models, the platform ensures accurate, real-time ASL translation and an engaging learning experience through tutorials, quizzes, and gesture validation. Its applications span education, healthcare, and customer service, promoting inclusivity and accessibility.

Future enhancements, including support for complex gestures, facial expressions, and offline functionality, will make the platform more robust and versatile. Incorporating gamified features, augmented reality, and support for additional sign languages will extend its global reach, making ASL education more interactive and inclusive.

By continually evolving with technological advancements, "SignSerenade" aims to bridge communication barriers, empower the Deaf community, and foster meaningful connections, contributing to a more inclusive and equitable society.

XII. FUTURE WORK

The future scope of "SignSerenade" focuses on enhancing its accuracy, scalability, and accessibility. To achieve better recognition of complex gestures, the integration of advanced neural network architectures and expanded datasets, including diverse signs and variations, will be prioritized. Incorporating facial expressions and contextual cues will further improve translation accuracy. Offline functionality through optimization for mobile and edge devices will make the platform accessible in low-resource environments. The learning module can evolve with gamified features, augmented reality-based gesture practice, and personalized pathways to make ASL education more engaging. Additionally, the modular design will enable the inclusion of other regional sign languages, broadening the platform's global impact and fostering inclusivity across cultural and linguistic boundaries.

XIII. REFERENCES

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